

Skin Care Products

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01

SKINCARE OVERVIEW

A brief overview of the skincare industry

OVERVIEW

Skincare is an essential part of people's daily routines, designed to maintain and improve the condition of the skin. It includes a wide range of products, such as cleansers, toners, lotions, oils, serums, and sunscreens.





02

DATASETS

Background of skincare datasets

DATASETS



PRODUCTS

Over 8k products from the Sephora online store



REVIEWS

About 1 million reviews on over 2,000 products



INGREDIENTS

Information on cosmetic substances and ingredients



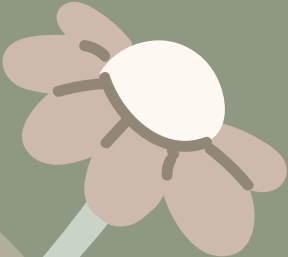
DISEASES

Dermatologic diseases from the A OCD



PICTURES

19,500 images of 23 types of skin diseases



03

CHALLENGE

Problem we faced transforming data



OUR CHALLENGES

STANDARDIZATION



01

TOKEN LENGTH



02

FORMATTING PROBLEMS



03

PROCESSING TIME



04

Solution



Our process for resolving each challenge





GEMINI INPUT LIMITS

GEMINI INPUT LIMITS

Standardization
Inputs

260,000

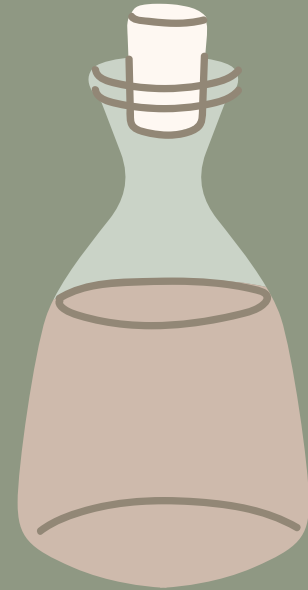
Gemini Character Limit

6,000,000

Product Ingredient Characters

1,000,000

Cosmetic Ingredient Characters



BATCH PROCESSING

Product Ingredients
(Unofficial Names)

100 Batches

10,000

Characters per batch

Cosmetic Ingredients
(Official Names)

400 Batches

15,000

Characters per batch



INVALID JSON OUTPUT

PRE-CLEAN NAMES

Original Names —→ Cleaned Names —→ Standard Names

01

".alpha.-methyl"

02

alphamethyl

03

alpha methyl



LONG RUN TIME

The top corners of the slide are decorated with stylized, light green foliage. On the left, there are leaves and a branch. On the right, there are more leaves and a circular element containing a leafy branch.

LONG RUN TIME

72 Hours

That's a lot of hours...

CONCURRENT PROCESSING

WORKER #1

Processing
Official Chunk
1/100

WORKER #2

Processing
Official Chunk
2/100

WORKER #3

Processing
Official Chunk
3/100

WORKER #4

Processing
Official Chunk
4/100

WORKER #5

Processing
Official Chunk
5/100

(BigQuery) Ingredient Replacements

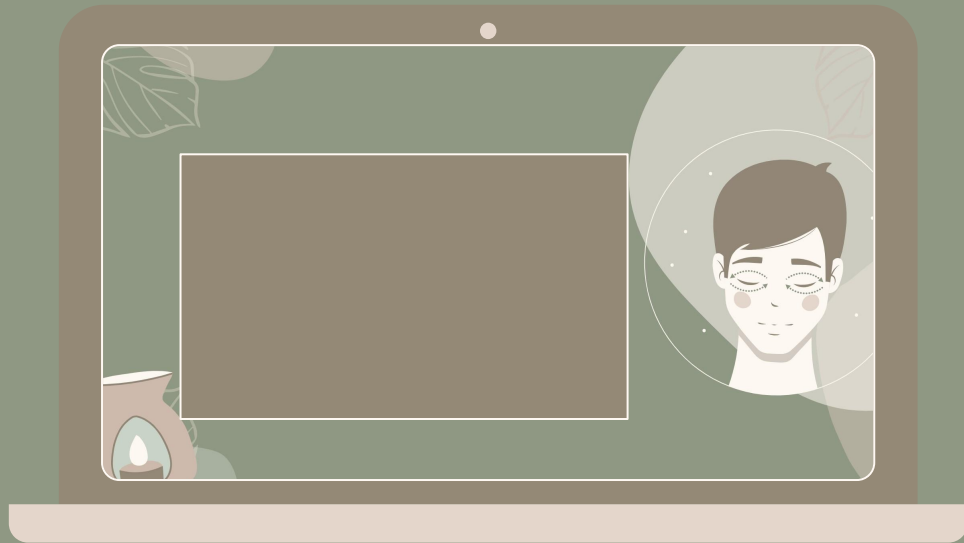
06

DEMO

Walk through of an interesting challenge



PRODUCT DEMO



THANKS!

